

Carve, Then Compose: Training-Free Domain Pruning and Recombination of Experts in Mixture-of-Experts LLMs

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July 30, 2026

Abstract

Expert pruning of Mixture-of-Experts (MoE) language models is believed to be safe only up to roughly 25% without retraining. We show that this ceiling is an artifact of demanding *general* capability, and that when only one domain must survive, an off-the-shelf fine-grained MoE tolerates far deeper cuts. On Qwen3-30B-A3B (128 experts/layer, top-8), a norm-aware contribution score computed from a teacher-forced prefill over only ~ 30 domain demonstrations retains 97% of baseline HumanEval pass@1 with 50% of routed experts removed, and 90% with 65% removed—while GSM8K collapses from 99.3% to 33.3%, confirming genuinely domain-targeted compression. A size-matched random control scores 26.8% HumanEval at the 50% ratio, an advantage of 62 points for informed selection. We further show that domain-pruned variants *compose*: the per-layer union of independently derived coding and math expert sets (52% of experts) scores within 1–2 points of the full model on both HumanEval (90.9%) and GSM8K (97.3%), while dropping 34 points on a non-STEM MMLU probe—a two-skill specialist assembled post hoc with no retraining. Composition holds for distant domain pairs (code+humanities: 84.1% HumanEval, 74.3% MMLU-humanities), whose keep-sets overlap less than sibling pairs. Finally, we observe a sharp asymmetry: procedural domains (coding, math) prune cleanly, but a knowledge domain (humanities) does not—the humanities-only specialist collapses even on its own domain (28.7% vs. 84.7% baseline), yet unions restore it to 74.3%. Our results suggest fine-grained MoE models are post-hoc composable at the expert level, and motivate applying the recipe to the most fine-grained public MoE, Kimi K3 (896 experts/layer).

1 Introduction

Sparse Mixture-of-Experts architectures dominate the largest open language models, but their checkpoints are monolithic: a user who needs only coding ability must nevertheless store and serve every expert, including those that exist to support poetry, law, and the long tail of world knowledge. One-shot expert pruning offers relief, but published results converge on a ceiling near 25% of experts before general performance degrades [1, 2, 3], with $\sim 50\%$ reachable only with retraining or distillation [4].

We ask a different question: **how much of the expert pool is needed if the user cares about a single domain?** This reframing removes the constraint that produced the 25% ceiling—capability outside the target domain is deliberately sacrificed rather than preserved. Building on few-shot expert localization [5], we prune an off-the-shelf fine-grained MoE with a simple training-

free pipeline and measure the full quality-versus-ratio curve, with controls the one-shot pruning literature typically omits.

Our contributions:

1. **A domain-restricted pruning curve far above the general ceiling.** With a norm-aware contribution score over ~ 30 demonstrations plus protection of “super experts” identified on a general set, coding capability survives 50% expert removal at 89.0% HumanEval pass@1 (baseline 91.5%) and 65% removal at 82.3%. The curve holds a plateau to its knee at $\sim 65\%$, decays steadily to 80% (45.1%), and collapses by 87.5%.
2. **Matched random controls quantifying the value of selection.** At identical keep-set sizes, random expert selection scores 26.8% vs. 89.0% (50% ratio) and 0.6% vs. 82.3% (65% ratio). Informed selection—not redundancy—carries the result.
3. **Training-free composition of domain specialists.** The per-layer union of independently derived coding and math keep-sets (67/128 experts, 75% of the sum of the parts) scores 90.9% HumanEval and 97.3% GSM8K—above both specialists and within 2 points of the unpruned model—while a non-STEM MMLU probe confirms it remains a specialist (50.3% vs. 84.7% baseline). A distant pair (codeUhumanities) also composes.
4. **A skills/knowledge asymmetry.** Identical machinery that yields strong coding and math specialists fails to produce a humanities specialist (28.7% on MMLU-humanities vs. 84.7% baseline), consistent with reports that semantic knowledge is distributed while procedural skill is localized [8, 9]. Unions partially restore the lost capability (74.3%), indicating the deficit is complementary rather than absolute.

All experiments are one-shot (no gradient updates), run on a single NVIDIA B200 GPU, and use masking that is exactly output-equivalent to checkpoint surgery (Section 3.4), making the study reproducible at minimal cost.

2 Related Work

General expert pruning. HEAPr [1] and MoNE [2] achieve near-lossless one-shot pruning at 20–25% on DeepSeek/Qwen MoE families; MoNE reports ~ 24 –25% as the maximum within 1% loss for Qwen3-30B-A3B—the same base model we use, providing a direct published anchor. MoE-I² [3] and MoE-Pruner [4] reach $\sim 50\%$ only with intra-expert decomposition, fine-tuning, or expert-wise distillation. Scoring criteria for one-shot pruning are surveyed in [15]; AIMER [14] removes calibration dependence in the task-agnostic setting. Our setting differs: we do not preserve general capability.

Domain-specific pruning. EASY-EP [5] introduced few-shot expert localization with output-aware importance, which our contribution score follows in spirit. C-PRUNE [6] clusters experts for task-specific compression in the biomedical domain, and recent work reports coding-domain pruning at $\sim 50\%$ on earlier-generation MoEs [7]. Relative to this line we contribute (i) the full quality-vs-ratio curve with its knee, (ii) matched random controls, and (iii) the composition results, none of which appear in prior domain-pruning studies to our knowledge.

Expert merging and modularity. An alternative to deletion is merging similar experts [12, 11]; REAP [11] finds pruning superior for one-shot compression. EMO [13] pretrains MoEs so that task-specific expert subsets can be selected at deployment, arguing that standard MoEs specialize syntactically rather than semantically. Our results provide a counterpoint: a standard, conventionally trained MoE already supports post-hoc subset selection and recombination for procedural domains with no pretraining modification—though EMO-style training may be what closes the gap we observe on knowledge domains. Analyses of expert specialization [8, 9, 10] motivate our protection of disproportionately active super experts.

3 Method

Given a routed-MoE checkpoint with E experts per layer and top- k routing, the pipeline has four training-free stages.

3.1 Routing capture

We run a teacher-forced prefill (no generation) over a small calibration set; each item is a prompt concatenated with a reference solution. Hooks on each layer’s router record, per token, the top- k expert indices and their renormalized gate weights, and per expert the L_2 norm of its output. Only dense per-(layer, expert) accumulators are stored.

3.2 Norm-aware contribution scoring

Routing frequency conflates syntactic regularity with importance, so we score expert e in layer l as the accumulated gate-weighted output norm

$$\text{score}(l, e) = \sum_t g_{l,e}(x_t) \|f_{l,e}(x_t)\|_2, \quad (1)$$

over the domain calibration set—the contribution the expert actually adds to the residual stream, following output-aware importance [5].

3.3 Super-expert protection

A small set of experts does disproportionate work regardless of domain [9]. On a *general* calibration set we flag experts whose gate mass exceeds $8\times$ the layer median, or that fall in the top 2% of a layer’s mass, and include them in every keep-set (6.5 experts/layer for our model). Domain pruning must not delete generic infrastructure.

3.4 Pruning, masking, and composition

A *manifest* lists surviving experts per layer: the protected set plus the top domain-scored experts up to the target ratio, with a floor of $2k$ experts per layer. For evaluation we *mask* rather than rewrite: a hook sets pruned experts’ router logits to $-\infty$, which is exactly output-equivalent to deleting the expert tensors and slicing the router (verified to $< 10^{-5}$ relative error on a synthetic model with planted structure). Masking makes a sweep cost zero storage; deployment uses real checkpoint surgery with identical outputs.

Composition is the per-layer set union of two manifests. No scores are recomputed and no calibration is rerun; the union inherits both protected sets.

4 Experimental Setup

Model. Qwen3-30B-A3B-Instruct-2507: 48 layers, 128 routed experts per layer, top-8 routing with renormalization, bf16, evaluated on one NVIDIA B200 with greedy decoding throughout.

Calibration sets. Authored for this study; no benchmark items are reused. Coding: 30 prompt-and-solution items across eight languages (roughly 15k tokens, teacher-forced). Math: 28 worked word and algebra problems, disjoint from GSM8K. Humanities: 24 essay-style answers spanning history, philosophy, law, and the arts. General: 16 mixed items, used only for super-expert protection.

Benchmarks. HumanEval pass@1 (164 problems, code-fence extraction, sandboxed subprocess execution); GSM8K exact-match on a fixed 150-problem subset; and a knowledge probe of 300 questions sampled from six non-STEM MMLU subjects (philosophy, world religions, jurisprudence, European history, marketing, prehistory). Baselines: 91.5% HumanEval, 99.3% GSM8K, 84.7% MMLU-hum. At these sample sizes, differences under ~ 2 –3 points are within noise; we interpret only larger gaps.

Ratios. $\{25, 50, 65, 70, 75, 80, 87.5\}$ % of routed experts removed. The 87.5% point coincides with the $2k$ -per-layer floor (16 kept, 8 active).

5 Results

5.1 Domain-restricted pruning beats the general ceiling

Experts removed	Kept/layer	HumanEval	GSM8K
0%	128	91.5	99.3
25%	96	91.5	97.3
50%	64	89.0	94.0
65%	45	82.3	33.3
70%	38	70.7	6.0
75%	32	63.4	2.0
80%	26	45.1	0.0
87.5%	16	0.0	0.0

Table 1: Coding-domain pruning sweep. In-domain (HumanEval) capability holds a plateau to $\sim 65\%$ while out-of-domain (GSM8K) collapses much earlier.

Coding-targeted pruning holds HumanEval within 2.5 points at 50% removal—double the published general-pruning ceiling for this model [2]—and within 10 points at 65%, where GSM8K has already fallen from 99.3 to 33.3. The in-domain curve is not smooth to the end: a plateau to the knee at $\sim 65\%$, steady decay through 80%, then collapse. The 87.5% point sits at the routing-slack floor and likely reflects that floor as much as the selection method.

Routing statistics foreshadow the curve: on the general set, mean Gini of per-layer expert mass is 0.63, $\sim 19/128$ experts carry 50% of layer mass, and $\sim 65/128$ carry 95%—half the pool shares the last 5% of routing mass.

5.2 Selection, not redundancy: matched random controls

Keep policy	Kept/layer	HumanEval	GSM8K	MMLU-hum
domain-scored (50%)	64	89.0	94.0	—
random (50%)	64	26.8	40.0	—
domain-scored (65%)	45	82.3	33.3	—
random (65%)	45	0.6	16.0	—
union code $\cup$ math	67	90.9	97.3	50.3
random size-matched	67	58.5	37.3	62.3

Table 2: Random selection at matched keep-set sizes. Informed selection wins in-domain by 32–82 points, but *loses* on the neutral domain.

Random selection at matched sizes loses 62 points of HumanEval at the 50% ratio and 82 points at 65%. At 67/layer the pattern inverts on the neutral domain: random *beats* the union on MMLU-hum (62.3 vs. 50.3) while trailing by 32–60 points in-domain. Random preserves diffuse mediocrity; informed selection concentrates capability into the chosen domains. (Random keep-sets here do not receive super-expert protection; isolating protection from domain scoring is left as an ablation.)

5.3 Domain specialists compose post hoc

Variant	Kept/layer	HumanEval	GSM8K	MMLU-hum
code-only (65%)	45	82.3	33.3	—
math-only (65%)	45	0.0	93.3	—
code$\cup$math	67	90.9	97.3	50.3
hum-only (65%)	45	0.0	0.7	28.7
code$\cup$hum	71	84.1	32.7	74.3
baseline	128	91.5	99.3	84.7

Table 3: Composition by per-layer set union. Unions exceed their parts in-domain while remaining specialists on the neutral probe.

The code and math keep-sets share 22.6 experts/layer, so their union is 67 rather than 90 (75% of the sum of parts). The union exceeds *both* specialists on their own domains and lands within 1–2 points of the unpruned model on both benchmarks at 52% of the experts, while its 34-point deficit on the non-STEM probe confirms it remains a two-domain specialist rather than a recovered generalist. Composition is not limited to sibling domains: code and humanities keep-sets overlap less (19/layer), and their union still scores 84.1 HumanEval and 74.3 MMLU-hum while GSM8K—covered by neither set—stays at specialist level (32.7).

The specialists are strikingly disjoint in capability despite sharing half their experts: the math-only model solves zero HumanEval problems, while the code-only model retains residual GSM8K (33.3), plausibly via shared arithmetic-adjacent machinery.

5.4 Skills prune; knowledge does not

The same pipeline that produces strong coding (82.3) and math (93.3) specialists at the 65% ratio *fails* for humanities: the hum-only variant scores 28.7 on its own domain, barely above the 25%

multiple-choice floor. This is consistent with analyses finding procedural competence concentrated in few experts while semantic knowledge is spread across many [8, 9]. Notably, the codeUhum union recovers humanities to 74.3—capability neither part exhibits alone, with code-only presumably contributing generic instruction-following and composition machinery. Two implications follow: (i) domain-restricted compression is substantially more effective for skill-like than knowledge-like domains at matched ratios; (ii) deficits from aggressive pruning can be complementary, and unions repair them cheaply.

We flag one confound: the humanities calibration set is essay-style while the probe is multiple-choice, so part of the hum-only collapse may reflect format sensitivity. A format-matched ablation is future work.

6 Limitations

Single model, single size, and a single seed per point; benchmark sample sizes give ± 2 –3 point noise. HumanEval and GSM8K are older benchmarks with likely training contamination, so our claims rest on *relative* retention under identical decoding rather than absolute scores. Masking realizes quality equivalence but not memory or latency savings, which require the mechanically verified surgery path. The 87.5% point is confounded by the routing-slack floor. Random controls omit super-expert protection. EMO [13] achieves related composability via modified pretraining; we do not claim post-hoc selection matches purpose-trained modularity, only that useful composability already exists in standard checkpoints.

7 Future Work

The natural target is Kimi K3 (896 experts/layer, top-16, 2.8T parameters, released July 2026)— $7\times$ finer expert granularity than studied here. If specialization grows with granularity [5], the knee should move right, and domain-restricted compression of the strongest open model would yield deployable specialists in the 200–400 GB range from a 1.56 TB checkpoint. Further directions: knowledge-domain pruning with format-matched calibration, per-layer adaptive ratios, expert-wise distillation to recover the post-knee region, and N -way unions (“models à la carte”).

Acknowledgments

Computations were performed using university research computing resources.

Reproducibility

Code, calibration sets, pruning manifests, and raw result files are available at <https://github.com/YashSarwaiya/carve-then-compose>.

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